Dr. Samuel Lee Jackson

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ABOUT

I am an astronomer & planetary scientist passionate about solving problems related to space safety. I have worked in roles covering: the characterisation of objects in space such as satellites and asteroids, planetary defence, and laboratory-based study of extraterrestrial samples (such as meteorites). These roles are all underpinned by my passion for ensuring a safe and sustainable space environment.

EXPERIENCE

Sybilla Technologies — **Remote, Milton Keynes**
Senior Sensor Network Operations Specialist — *2024 - present*

- Leading technical delivery of satellite characterisation capabilities from ground-based optical sensors (hardware and software)
- Maintenance and operations of a global network of space surveillance sensors
- Ensuring high-quality, timely space surveillance data delivery to customers and partners
- Development of algorithms & software (mostly in Python) for enhancements to space surveillance capabilities
- Development and delivery of internal and external training programs related to space safety and associated technologies

The Open University — **On-Site, Milton Keynes**
Post-Doctoral Research Associate (Asteroid Surfaces) — *2023 - 2024*

- Thermophysical modelling (C++/CUDA) of asteroids to support surface property characterisation
- Robotic telescope observations to determine asteroid physical properties
- Thermophysical analysis supporting ESA's Hera mission and NASA's Lucy mission
- Spectroscopic characterisation of meteorites for parent-body identification, supporting new understandings of Solar System formation and evolution

University of Edinburgh — **Remote, Milton Keynes**
Post-Doctoral Research Associate (Part-Time, 6 month contract) — *2022 - 2023*

- Time-of-impact observations from Kenya as part of NASA's DART mission to crash a spacecraft into an asteroid to re-direct it in the world's first planetary defence test
- Characterisation of impact properties and plume evolution
- Advisory role in a project to develop the first permanent optical telescope in Kenya

The Open University — **On-Site, Milton Keynes**
Emergency Support Astronomer/Observatory Software Consultant — *2019 - 2022 (Multiple Contracts)*

- Ensuring safe operation of two robotic telescopes during remote use by university students
- On-call emergency support and remote/in-person maintenance of observatory systems
- Training for students and staff on observatory operations and maintenance
- Development of data delivery scripts and processes for student data

AWARDS AND MEMBERSHIPS

- Member of the NASA Lucy mission Surface Composition Working Group (2023 - 2025)
- Member of the ESA Hera mission investigation team (2023 - 2024)
- Asteroid 30226 Samuelleejackson (2000 GY137) named in recognition of my work in planetary astronomy and robotic telescope observations (2023)
- Member of the NASA DART mission investigation team (2022 - 2023)
- Rotary Prize for the Faculty of Sciences, University of Kent (2019)
- Dean's Prize for the Faculty of Sciences, University of Kent (2018)
- Fellow of the Royal Astronomical Society (2017 - Current)

EDUCATION

PhD (Astronomy & Planetary Science) **Milton Keynes**
The Open University *2019 - 2023*

Thesis Title: Characterising Asteroid Spin and Surface Properties using Small-Aperture Telescopes

- Collected over 1,000 hours of observations with the OpenScience Observatories to determine physical properties of 18 asteroids
- Characterised camera and whole-system behaviour through long-term studies of calibration and science frames, reducing noise in the final calibrated images and optimising data collection strategy to avoid mechanical issues such as wind-shake
- Evaluated the ability of small telescopes to contribute high-quality data of faint, moving targets such as asteroids
- Investigated the viewing geometry dependence in near-Earth asteroid phase curves, assessing how this systematic uncertainty propagates through to further study of these objects
- Performed remote and on-site maintenance of the OpenScience Observatories, contributing to testing and characterisation campaigns after upgrades while liaising with hardware and software suppliers to quickly resolve issues

MPhys (Astronomy, Space Science, and Astrophysics - First Class Honours) **Canterbury**
University of Kent *2015 - 2019*

Dissertation Title: Asteroid (1917) Cuyo: The Formation of a Binary Asteroid System?

- Processing of data from telescopes around the world to validate the presence of long-term changes in the spin-state of a target asteroid
- Participated in observing evenings using the robotic Beacon Observatory, gaining familiarity with telescope control software such as MaxIm DL
- Modules covering astronomical data analysis, spacecraft systems engineering, astrophysics and numerical methods

GCSE & A-Level Education **Westcliff-on-Sea**
Westcliff High School for Boys *2008 - 2015*

KEY SKILLS

- Programming: Python, C++, CUDA
- Experienced with Bash, familiar with Powershell
- Operating systems: Linux, Windows
- Telescope observations, operations, troubleshooting and maintenance
- Data analysis and statistics
- Public speaking across a wide range of audiences
- Writing technical documentation
- Collaboration as part of small and large international teams

SELECTED PEER-REVIEWED PUBLICATIONS

- **Jackson, S. L.**, Kolb, U. C., & Green, S. F. (2021). Asteroid Photometry with PIRATE: Optimizations and Techniques for Small-Aperture Telescopes. *PASP*, 133, 075003
- **Jackson, S. L.**, Rozitis, B., Dover, L. R., Green, S. F., Kolb, U. C., Andrews, A. E., & Lowry, S. C. (2022). The Effect of Aspect Changes on Near-Earth Asteroid Phase Curves. *MNRAS*, 513(2), 3076
- Chabot, N., et al. **(incl. Jackson, S. L.)** (2024). Achievement of the Planetary Defense Investigations of the Double Asteroid Redirection Test (DART) Mission. *PSJ*, 5(2), 49
- Rozitis, B., Green, S. F., **Jackson, S. L.**, et al. (2024). Pre-impact Thermophysical Properties and the Yarkovsky Effect of NASA DART Target (65803) Didymos. *PSJ*, 5(3), 66
- Rider-Stokes, B. G., Stephant, A., Anand, M., Franchi, I. A., Zhao, X., White, L. F., Yamaguchi, A., Greenwood, R. C., **Jackson, S. L.** (2024). Evidence against water delivery by impacts within 10 million years of planetesimal formation. *EPSL*, 642, 118860
- **Jackson, S. L.**, Rozitis, B. (2024). Thermophysical modelling of eclipse and occultation events in binary asteroid systems. *MNRAS*, 534(3), 1827
- **Jackson, S. L.**, Emery, J. P., Rozitis, B., Christensen, P. R., Spencer, J. R., Mottola, S., Hamilton, V. E., Howett, C. J. A., Marchi, S., Noll, K. S., Levison, H. L. (2025). Thermal-IR Observations of (152830) Dinkinesh during the Lucy Mission Flyby. *PSJ*, 6(7), 168